

# SERVICE NOTES

Oct. 14, 1999

Page 1 of 9

FILE TITLE

Signa® RF Subsystem

SN 60916A

## ERBTEC AMPLIFIER SAFETY AND ERRATA

### Reprint

This Service Note has been updated and reprinted because we do not want to hear of injuries sustained by Field Engineers. A point was made after a recent incident noting the fact that there are many FEs that are in the process of being 'cross trained' in the MR modality, and may NOT be aware of the seriousness of potential personal injury while working on the Erbtec RF amplifiers.

Read this updated Service Note and the Attachment #1 at the end of this document. If you still have any doubts, get them cleared up **before** attempting to work on an Erbtec!

### Effectivity

All FE's servicing Signa MR systems with Erbtec, 5, 8, and 20 KW tube type amplifiers.

### Problem

- Potential shock hazard while working on Erbtec tube type amplifiers.
- It has been brought to our attention that several FEs in the field, both domestically and globally, have been shocked, while working on Erbtec tube type RF Amplifiers.

### Solution

The intent of this Service Note is to correct any misunderstandings one might have while working on the Erbtec amplifiers. It is also directed to the experienced FE who has been trained some time ago on the Signa system. It will not be watered down with referencing specific instructional material. It is assumed everyone will be using the latest CD ROM when necessary.

This note will address the proper, safe, way to shut the amp down, to Protect the FE from injury, and also Protect the amp from any potential damage. FEs coming through Ed Center classes still (occasionally) hear of FEs being shocked while working on the amps in the field. Hopefully, this Service Note, and use of the latest CD ROM will correct this problem.

Please read the attached pages, and if they are NOT fully understood, contact your Support personnel or the authors of this Service note for further explanation. There cannot be any 'gray areas' of understanding of these topics.

**They must be fully understood, or you should NOT be working on this equipment.**

# SERVICE NOTES

Oct. 14, 1999

Page 2 of 9

FILE TITLE

Signa® RF Subsystem

SN 60916A

## ERBTEC Power Down Preparation

### WARNING!

Simply turning the Erbtec circuit breaker off is NOT the safe way to protect yourself. Tests have repeatedly confirmed that hazardous voltages are still present several minutes after only turning the breaker off. If there were problems with the unit, these times could be considerably longer! Stored energy inside the Erbtec Amplifier has the potential of bodily damage and can be life threatening. Perform the following steps to ensure unsafe voltages are no longer present before servicing inside the Erbtec Amplifier.

1. Assumption - the covers are attached, as you are actively working on the amp.
2. With a serial link cable (46-320252P1) attached, laptop **COM 1** to Erbtec **J-2204** connector, command the amp to the '**OFF**' mode. (For laptop setup - see Attachment #2 at the end of this document.)
  - enter '**00**' (that's a letter O, and the number zero) (entry is not case sensitive)
  - In doing this, the microprocessor causes the tube(s) to be forward biased thus dissipating the stored energy in the power supply. This process takes about 2 to 3 seconds. Both the 3 and 6 kV power supplies should be brought to zero volts within seconds if nothing is wrong with the amp. (This is your first protective step.)
3. After the 2-3 seconds have passed, turn the Erbtec circuit breaker off. (Your second protective step.)
4. Time now to apply LOTO. Remove and lock the AC power cord and wait 3 minutes as the warning label on the back of the Amp instructs you to do. (Your third protective step.)

# SERVICE NOTES

Oct. 14, 1999

Page 3 of 9

FILE TITLE

Signa® RF Subsystem

SN 60916A

## WARNING!

Use extreme caution in the next step as there still may be hazardous voltages present inside the Erbtec Amplifier capable of bodily harm!!! Follow closely the remaining steps to insure personal safety.

5. Remove the respective cover(s). With a DVM, set to the highest DC level, measure and look for any potential high voltage source within the amp, i.e. the HV resistor on the power supply; the surface of the PA and / or IPA (if applicable) tube(s) with respect to the Amp frame or chassis. Anywhere you may be expecting to put your hands. The amp power supplies should now be at zero volts. Remember though, you are working on this amp looking for a possible problem. Some component could be defective! You must understand and follow the next statements before putting your hands inside the unit!

- **If a measured potential is above 20 volts, something is definitely wrong within the amp. You have found something wrong! Proceed with extreme caution!**
- **Do NOT short the power supply to ground if above 20 volts. (NOTE: this may vary with some existing Erbtec documentation!) Wait whatever time is necessary for the voltage to drop below 20 volts.**

- first ; you don't want to be adding to possible existing problems.

- secondly; if there was nothing wrong with the supply before, there may be a problem induced if you short the supply out with too much power in it. Protective diodes D1 and / or D12 may be blown. (By the way, these diodes are not FRUs.)

- and finally; also at peril are the integrity of diodes on the PA and / or IPA Input boards. And you don't want to be changing them out unless absolutely necessary.

6. When the measured potentials are below 20 volts, short out the respective 6 KV and 3 KV (if present) power supply(ies) with a 1/8 x 6" screwdriver as called out in your latest CD ROM.
7. **If you do not do the following, this is where you most often expose yourselves to danger.**

**Immediately after shorting the supplies to ground, attach a grounding wire to the amp frame / chassis first, then attach the opposite end to the power supply. (Don't wait 3 to 5 minutes to do this. Do it Immediately!)** Depending on where you are troubleshooting, this can

# SERVICE NOTES

Oct. 14, 1999

Page 4 of 9

---

**FILE TITLE**

Signa® RF Subsystem

SN 60916A

---

be the power supply standoffs on the supply; the tube cases (both PA and IPA); anywhere! Two grounding cables are required for the 20 KW amp, one each for the 3 and 6 KV supply. One cable only is required for the 5 or 8 KW amp, with only the 6 KV supply present.

- The intent of the grounding strap(s) is to prevent any buildup of power in the caps in the power supply. This buildup could be hazardous because if you THINK the supply is safe, then suddenly, you get a small shock, you most probably will remove your hands quickly, risking injury of scraping your hands etc.
- Final note on the grounding strap, or wire. It does NOT have to be a large gauge wire. Once the supply has been grounded, it only takes a very small gauge wire to keep it at ground potential.

8. The amp should now be safe and your only other caution would be to use proper ESD practices while working inside the amp. Happy (SAFE) troubleshooting.

# SERVICE NOTES

Oct. 14, 1999

Page 5 of 9

FILE TITLE

Signa® RF Subsystem

SN 60916A

## Additional Erbttec Errata

- Pressure Sensing Transducer



**NEVER bypass the transducer for the purpose of gaining access to the VTAC (Vacuum Tube Amplifier Cavity).** This erroneous practice will do nothing more than subject you to potentially 'life threatening' high voltages and also drastically shorten the life of the PA and IPA tubes! **Do not access the VTAC with the amp under power!** If the Transducer is a problem, fix the problem.

Student comments routinely reference having been told to "bypass the pressure sensing transducer if certain problems arise. Primarily if you want to gain access to the VTAC.

- **This statement is wrong!** If you were told this in the past, we want to correct your understanding as of now.

First, you want to understand what function the transducer performs. It is there to measure a positive air pressure inside the VTAC. This is to insure proper cooling to the tubes and related circuitry. If you should ever receive a service call and, while troubleshooting, an error message points you to an "88 - Fan air pressure insufficient" fault or an 89 fault, the first thing you would want to do is to check the squirrel cage blower motor to insure it is properly rotating. High speed! If it appears to be at full speed, (under power, with only the left side cover off!) the next thing to do would be to (safely) shut the unit fully off, and remove the blower motor. Remove the buildup of dirt, dust, and lint that has accumulated on the wave guide, (or popularly referred to as 'the honeycomb'). Reinstall the blower motor and retest.

Typically, this will eliminate the root cause of the errors because the pressure will again be positive inside the VTAC. It is also recommended that you remember this and possibly check other units as a preventive measure. Remember it also in a couple more years. That's how long it typically takes to clog up again.

- The biggest point to remember is to **NEVER** bypass the pressure sensing transducer. **Fix the air flow problem!**

# SERVICE NOTES

Oct. 14, 1999

Page 6 of 9

**FILE TITLE**

Signa® RF Subsystem

SN 60916A

## **Some extra information on the pressure sensing transducers.**

- NEVER adjust the small screw on the transducer. It has been factory set to respond to specific pressures. If you have adjusted this screw on units, please take time to reorder the transducer and replace it. You have not fixed the problem; you have potentially shortened the life of the PA and IPA tubes because they may not be receiving sufficient cooling.
- If you truly suspect the transducer is defective, look in the lower front area of the power supply. In most cases, you will find another transducer just like the one used on the VTAC. This one however is not used. Swap them to see if it resolves the problem (permanently).
- Final comment. In the 9 plus years we have had our 5 or 6 Erbtects around the Education Center, we have not had any pressure sensing transducer failures. In all cases, pressure problems were related to dirt / dust buildup on the waveguide.

Good luck. And let's not hear any horror stories about FEs being injured while working on the RF amps.

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# SERVICE NOTES

Oct. 14, 1999

Page 7 of 9

FILE TITLE

Signa® RF Subsystem

SN 60916A

## Attachment : #1

This is an actual Email from a very experienced Field Engineer. Please read it and review the Service Note again if you have any questions. By the way, follow-up with this individual revealed that the amp in question "Was T3'd! Returned! We didn't want to take any other chances with it!"

As you may already know, we recently had an accident here in **xxxxxx** where an FE was shocked by an RF amp, causing burns to his fingers and hand, leading to an overnight stay in the hospital. I wanted to send you guys a message summarizing the events and pointing out what I believe to be a very important point.

The system being worked on was an **xxxxx** mobile .5T 5.X. The RF amp was out of service. I arrived Monday morning to begin working and started with a quick survey of the situation to get a decent understanding of the problem. As a part of this process, I had my laptop hooked up to the serial port on the back of the amp and had been trying to get it to do things it would not do (i.e. pass AFT and go to ready).

Here begins the sequence of events leading to the accident.....I commanded the amp to standby and then to off, both commands were executed properly by the amp. After about 5 mins. the breaker on the back of the RF amp was turned off. I was preparing to do some work on the inside of the amp, so before going any further I wanted to verify that all voltages were gone. I was kneeling on the PA tube side of the amp, the amp was out on its rails and the side cover was off. I asked the FE working with me to hand me my DVM, which he did. After he passed me my DVM over the top of the amp - remember it was extended from the cabinet - he rested his arms on the top of the amp. His right hand was allowed to hang over the side of the amp where it made contact with the top of the PA tube, creating a path to ground. A serious shock resulted. He spent one night in the hospital and has since fully recovered.

NOW, the important point (at least from my perspective) Commanding the amp to off is *supposed* to bleed off all high voltage at the PA tube within a few seconds; several minutes had passed. Likewise, turning off the amp main breaker should bleed off high voltage, although not nearly as fast. **Point being: the amp should have been safe but it was NOT.**

I literally had leads in hand and was only a few seconds from checking exposed HV areas on the tube when the accident occurred. Had I made it that far, an accident could have been avoided. SO.....

PLEASE learn from our incident. I look at it this way: what if it was 1:00 am on a Saturday, you were by yourself and confined to the 3 foot by 3 foot space created when the amp is extended on the service rails. And you found the same 6KV with your hand, and it coursed through your body to ground through your knees. If the shock didn't kill you, trying to fight your way out of the confined area could really hurt.

One last thing, **I do use my LOTO procedures, but did not during this time.** Why??

Because I was 3 feet from the breaker and I knew nobody was going to come in and turn it on. Would it have made any difference had I used LOTO? No, the problem was not power still being applied, rather "stored energy". Again, please verify, it really could save your life.

# SERVICE NOTES

Oct. 14, 1999

Page 8 of 9

---

**FILE TITLE**

Signa® RF Subsystem

SN 60916A

---

## Attachment: #2

How do I set up my laptop to communicate with the Erbttec?

### Terminal Setup

1. On your laptop, lower left side, select **START**, then **RUN**.
2. Type **TERMINAL**, then click **OK**.
3. Select **SETTINGS**, then **TERMINAL EMULATION**, then **DEC VT100(ANSI)**, then **OK**.
4. Back to **SETTINGS**, and select **COMMUNICATIONS**.
5. From the drop down window, set up as follows:
  - Connector           **COM 1**
  - Baud Rate           **9600**
  - Data Bits           **8**
  - Stop Bits           **1**
  - Parity               **none**
  - Flow Control       **Xon/Xoff**
6. Select and click on **OK** to exit the menu.
7. You should now be able to communicate with the Erbttec.
  - You also have to remember, if you're working on an older (pre 5.5) system, you have to disable system Bi-cycle Link control of the Erbttec. Refer to your current CD for that process.
8. When finished, select **FILE**, and **EXIT**. The laptop will now ask if you want to save these changes for future use. The suggestion is to answer **YES**; then name the file, for example, **ERBTEC.TRM**. (Now for later use, go to **START**, **RUN**, select or type **ERBTEC.TRM**, **OK**, and you're ready to go.)

### Hyper Terminal Setup

1. On your laptop, lower left side, select **START**, **PROGRAMS**, **ACCESSORIES**, and finally, **HYPER TERMINAL**.
2. Double click on the icon **HYPERTRM.EXE**.
3. Enter a name and select an icon for your setup, then **OK**.
4. Select **CONNECT USING**, then **DIRECT TO COM 1**, then **OK**.

# SERVICE NOTES

Oct. 14, 1999

Page 9 of 9

---

**FILE TITLE**

Signa® RF Subsystem

SN 60916A

---

5. This brings up COM1 PROPERTIES, to which you input the following:
  - BITS PER SECOND               **9600**
  - DATA BITS                       **8**
  - PARITY                           **NONE**
  - STOP BITS                       **1**
  - FLOW CONTROL               **XON/XOFF**
  - finally                           **OK**
6. You should now be able to communicate with the Erbtec.
  - You also have to remember, if you're working on an older (pre 5.5) system, you have to disable system Bi-cycle Link control of the Erbtec. Refer to your current CD for that process.
7. When finished, select **FILE**, and **EXIT**. The laptop will now ask if you want to save these changes for future use. The suggestion is to answer **YES**; then name the file, for example, **ERBTEC.TRM**.
8. Now for later use, go to **START**, **PROGRAMS**, **ACCESSORIES**, and finally, **HYPER TERMINAL**. If you saved the info, you should now see an icon with your title for **ERBTEC.TRM**. Click on it and you are ready to go again.)